

Literature Review: Predicting 30-Day Hospital Readmission from Noisy Diabetes Records in US Hospitals

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I. INTRODUCTION

Hospital readmission is an important clinical outcome because it reflects both patient health status and the quality of care provided after discharge. In diabetes care, predicting readmission is especially challenging because hospital records are often noisy, incomplete, and highly heterogeneous.

In this section, four studies previously conducted on diabetic patients' readmission are reviewed to understand how machine learning has been applied, what kinds of models have been used, how noisy records were handled, and what limitations remain in these works.

II. PAST WORK

The study by Ameen Abdel Hai et al. [1] introduces a comparative framework for predicting unplanned hospital readmission among diabetes patients by combining deep learning and traditional machine learning models. The study utilizes a large EHR-based dataset structured under a common data model, where patient information is represented as sequential encounter data with up to 50 unified features per visit.

For evaluation, an LSTM model is compared with several conventional approaches, including Random Forest (RF), Logistic Regression (LR), Multilayer Perceptron (MLP), AdaBoost, Decision Tree, Support Vector Machine (linear and RBF kernels), K-Nearest Neighbors (KNN), and Naïve Bayes.

The findings indicate that LSTM achieves the highest overall performance, with an AUROC of 0.79, recall of 0.81, F1-score of 0.80, and accuracy of 0.81, surpassing the best-performing traditional model, Random Forest, which records an AUROC of 0.72.

The study further highlights the significance of temporal patient history, feature representation, and the inclusion of

laboratory data, while noting that performance is influenced by the number of prior encounters available. Overall, the results suggest that deep learning methods can offer improvements over traditional techniques for readmission prediction from heterogeneous clinical data, although the performance gains remain moderate.

Al Qahtani et al. [2] conduct a study focused on predicting hospital readmissions for diabetes patients using both machine learning and deep learning techniques. The work aims to support clinical decision-making, improve resource allocation, and reduce healthcare costs by addressing issues such as class imbalance and data heterogeneity.

To achieve this, preprocessing strategies including SMO-TENC for binary classification and Random Under-Sampling for multiclass scenarios are applied. Using the "Diabetes 130-US Hospitals" dataset, multiple models are assessed, including Logistic Regression, Decision Trees, Random Forest (RF), XGBoost, Deep Neural Networks, and a Multi-Stage Classifier.

The results demonstrate that Random Forest performs best in binary classification, achieving 94% accuracy and 0.97 AUC-ROC for ≤ 30 -day readmissions, while the Multi-Stage Classifier shows superior performance in multiclass prediction, with 80% accuracy and 0.89 AUC-ROC across no, early, and late readmission categories.

These findings emphasize the effectiveness of tailored data balancing techniques and staged classification in improving prediction reliability.

Auxilia Michael et al. [3] develop ReDiaSafe, a machine learning-based system designed to predict 30-day hospital readmission risk in diabetic patients. The objective of this system is to assist clinicians in identifying high-risk individuals

and providing personalized follow-up recommendations.

Similar to previous studies, the Diabetes 130-US Hospitals dataset is used, with preprocessing steps including feature analysis, categorical encoding, numerical scaling, and extraction of key predictors such as age, number of medications, time in hospital, insulin usage, and prior admissions.

The study evaluates several models, including LightGBM, Logistic Regression, Random Forest, and XGBoost, with LightGBM achieving the best overall performance. Specifically, it reaches 89.6% accuracy, 85.3% precision, 87.1% recall, 86.2% F1-score, and 0.91 AUC-ROC.

A notable contribution of this work is the integration of an interactive AI chatbot that provides individualized healthcare suggestions, enhancing the practical usability of the system for both patients and healthcare providers.

According to Li et al. [4], a machine learning-based framework is developed to predict 30-day hospital readmission in diabetic patients using electronic health record data. The study examines 101,766 inpatient encounters from the UCI Diabetes 130-US Hospitals dataset and applies preprocessing techniques such as feature imputation, scaling, and one-hot encoding before training the models using an 80/20 train-test split.

Four models, Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks, are compared in terms of performance. The results show that XGBoost achieves the best overall balance between predictive performance and interpretability, particularly in identifying readmitted cases.

Additionally, SHAP analysis is employed to determine the most influential features, including prior inpatient visits, number of medications, duration of hospital stay, and insulin-related variables. This interpretability component provides valuable clinical insights into how different factors contribute to readmission risk.

TABLE I
SUMMARY OF RELATED WORK

Authors	Title	Year	Model	Dataset	Accuracy / Key Result
Ameen Abdel Hai et al. [1]	Comparative framework for predicting unplanned hospital readmission among diabetes patients	2023	LSTM, RF, LR, MLP, AdaBoost, DT, SVM, KNN, NB	Large EHR-based dataset under common data model, sequential encounter information	LSTM: AUROC 0.79, Recall 0.81, F1 0.80, Accuracy 0.81; RF: AUROC 0.72
Al Qahtani et al. [2]	Predicting Hospital Readmission for Diabetes Patients	2025	LR, DT, RF, XGBoost, DNN, Multi-Stage Classifier	Diabetes 130-US Hospitals dataset	RF: 94% Accuracy, 0.97 AUC-ROC; Multi-Stage Classifier: 80% Accuracy, 0.89 AUC-ROC
Auxilia Michael et al. [3]	ReDiaSafe: A Novel Approach for Predicting 30-Day Diabetes Patient Readmission Risk	2024	LightGBM, LR, RF, XGBoost	Diabetes 130-US Hospitals dataset	LightGBM: 89.6% Accuracy, 85.3% Precision, 87.1% Recall, 86.2% F1, 0.91 AUC-ROC
Li et al. [4]	Comparison of machine learning models for predicting 30-day readmission rates for patients with diabetes	2024	LR, RF, XGBoost, DNN	UCI Diabetes 130-US Hospitals dataset	XGBoost and RF performed best overall; SHAP used for interpretability

III. COMPARATIVE ANALYSIS

All four studies focus on predicting hospital readmission in diabetes patients and share a common goal of improving early risk identification so that clinicians can intervene sooner and reduce avoidable readmissions. They rely heavily on electronic health records or UCI/Health Facts-style datasets and emphasize the importance of preprocessing, feature selection, and handling class imbalance or temporal structure. Common predictive features include prior admissions, medication use, length of stay, age, and laboratory results.

The main differences lie in the modeling strategies adopted across the studies. A multi-stage classification approach for multiclass prediction is introduced in [2], while a LightGBM-based framework with an integrated AI chatbot is presented in [3]. In contrast, Li et al. [4] focus on improving interpretability through SHAP analysis. A broader comparative framework is explored by Ameen Abdel Hai et al. [1], where LSTM is shown to outperform traditional models when temporal data is effectively utilized.

Preprocessing approaches also vary significantly. Extensive data cleaning, encoding, and balancing techniques are applied in [2], whereas a more lightweight preprocessing strategy is followed in [3]. A structured machine learning pipeline with validation is adopted in the study by Li et al. [4], while a sequence-based encounter representation is emphasized by Ameen Abdel Hai et al. [1].

Performance outcomes differ depending on the task. In [2], Random Forest achieves the best results for binary classification, while LightGBM performs best in [3]. Strong performance from both XGBoost and Random Forest is reported in [4], whereas LSTM demonstrates the highest overall performance in [1], although the improvement over traditional models remains moderate.

IV. RESEARCH GAPS

Despite being promising, the reviewed studies share several important limitations that indicate clear research gaps in diabetes readmission prediction. Firstly, most models rely on retrospective EHR data, making them sensitive to missing values, class imbalance, and limited generalizability.

The study by Al Qahtani et al. [2] highlights challenges in distinguishing readmission types. Auxilia Michael et al. [3] propose a chatbot system but lack real-world validation. Li et al. [4] improve interpretability but rely on a single dataset. In [1], it is shown that the effectiveness of deep learning models declines when patient history data is limited.

These gaps indicate the need for more robust, generalizable, and clinically deployable models.

To address this, the current study proposes a structured preprocessing pipeline: selecting first encounters, splitting data into training, validation, and test sets, imputing missing values using training-derived statistics, encoding categorical features, scaling numerical variables, and applying SMOTE only to the training set. This approach aims to reduce overfitting and improve generalizability.

V. CONCLUSION

Overall, the reviewed literature shows that machine learning and deep learning are promising for predicting hospital readmission in patients with diabetes, but existing studies still face challenges related to class imbalance, data quality, interpretability, and generalizability across healthcare settings. These limitations highlight the need for a more structured and clinically useful predictive framework. Therefore, the present study aims to develop a robust readmission prediction model for the Diabetes 130-US Hospitals dataset by applying careful preprocessing, balanced data handling, and systematic model evaluation to improve reliability and practical usefulness in real-world healthcare applications.

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